Project Brainstorming & Planning – June 1, 2026

Project Concept: Hybrid Diagnostics Application

The team decided to focus on a healthcare application designed to diagnose common human diseases (e.g., diabetes, cholesterol issues) using a combination of patient vitals and image processing.

- **Core Functionality:** Use patient data (glucose, cholesterol) and medical imagery (X-rays, retinal scans, MRI slices) to train an ML model for disease identification.
- **Workflow:**
    a. Collect patient vitals and medical images.
    b. Pre-process data using **OpenCV** (resizing images, handling missing values like BP).
    c. Train the ML model to predict diseases or suggest alternative possibilities based on symptoms.
- **Data Source:** Utilizing a large dataset (approx. 17,000 rows) from a government source.

Tech Stack

- **Data Handling/Analytics:** NumPy, SciPy, Pandas.
- **Computer Vision:** OpenCV.
- **Visualization:** Matplotlib/Seaborn (for predicting future symptom occurrences).
- **Machine Learning:** Scikit-learn (primary); TensorFlow/PyTorch (optional for deep learning/tumor identification).
- **Backend/Storage:** Initially using CSV or text files for storage to simplify the MVP; SQL may be integrated later.
- **Frontend:** HTML, CSS, JS, and REST APIs.

Roles & Responsibilities

- **Nagarithwik Vallala:** Visualization, Computer Vision (OpenCV), and Data Handling.
- **[Other Speaker]:** Machine Learning (Scikit-learn) and Data Analytics.
- **Vishant:** Frontend development and API management.
- **Database Management:** Currently unassigned; the team will skip complex SQL storage for now to focus on the core ML model.

Learning Plan & Timeline

The team aims to complete the project by **July 20th**, ahead of mid-semester exams starting August 17th.

Study Schedule for ML/Data Analytics:

- **Focus:** Chapters 3, 4, 5, 6, and 8 of the reference textbook (skip Chapters 1, 2, and 7; Chapter 9 is optional).
- **Daily Goal:** Approximately 3 hours of study per day.

Other Topics Discussed

- **Blockchain:** Considered for agro-based industries but discarded as it was a repeat of a previous semester's project.
- **AI Power Grids:** Discussed energy efficiency management but deemed too focused on embedded systems/IoT rather than data analytics.
- **Misinformation/Deepfakes:** Discussed identifying AI-generated content, but concerns were raised regarding the availability of high-quality training datasets and the high risk of failure.
- **Finance Transparency:** Briefly mentioned tracking large-scale government transactions, but determined it was not a good fit for a data analytics project.

Action Items

- Nagarithwik: Share the specific textbook chapters and resources with the team.
- [Other Speaker]: Begin studying Data Analytics (Chapters 3-5) and Machine Learning (Chapter 8).
- Team: Reach out to Vishant to confirm his availability for frontend/API work.
- Team: Hold weekly sync meetings to track progress.

Perfect, Sridhar — let's design a **complete project outline** that combines **OpenCV (for medical images)** with **ML (for patient data)** using your preferred Python stack. This way, you'll have a hybrid system that can handle both **numerical health records** and **visual diagnostics**.

🏥 Project Topic

AI-Powered Hybrid Diagnostics: Combining Patient Data + Medical Imaging

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⚙️ Tech Stack

- **Data Handling**: numpy, scipy, pandas
- **Visualization**: matplotlib
- **Machine Learning**: scikit-learn (Random Forest, Logistic Regression), optional TensorFlow/PyTorch for deep learning
- **Computer Vision**: opencv-python (OpenCV)
- **Optional:** scikit-image**, PIL for advanced image preprocessing**

📊 Workflow Outline

1. **Collect Data**
   - Tabular: Patient vitals (BP, glucose, cholesterol, age, BMI).
   - Imaging: X-rays, retinal scans, or MRI slices.
2. **Preprocessing**
   - Tabular: Handle missing values, normalize ranges.
   - Imaging: Resize, grayscale, denoise with OpenCV.
3. **Feature Engineering**
   - Tabular: Derived features (BMI, risk scores).
   - Imaging: Extract contours, textures, or CNN embeddings.
4. **Model Training**
   - Combine tabular + image features into a single dataset.
   - Train ML models (Random Forest, Gradient Boosting, or CNN).
5. **Evaluation**

    - Metrics: Accuracy, ROC-AUC, precision-recall.
    - Visualizations: Confusion matrix, risk distribution plots.
6. **Deployment (Optional)**
    - Build a simple Flask/Django web app for doctors to upload patient data + images.
    - Show predictions and risk scores with visual explanations.

🚀 Why This Project Works

- Tackles **real-world healthcare inequality**.
- Uses your **preferred stack** (numpy, scipy, pandas, matplotlib, ML, OpenCV).
- Academically strong: integrates **data science + computer vision**.
- Scalable: can expand into a full diagnostic assistant.
